LUMA

Local Unit Modulus Assignment

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This tool was developed in the IMSB group at the Insigneo Institute at the University of Sheffield, UK. Please visit the Insigneo Institute GitHub page for more information on other projects and research.

If you use this software in your research, please cite the LUMA citation file.

The tool is heavily inspired by previous work done by:

- Elise Pegg and Richie Gill [1, 2].
- Istituto Ortopedico Rizzoli [3, 4, 5, 6, 7, 8].
  - Bonemat
  - ALBA (Agile Library for Biomedical Aplications)

For more information, please see the LUMA Documentation.

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Overview

LUMA is a rust tool, designed to assign bone material properties to finite element meshes based on CT scan data. It provides:

- **High Performance**: Sub 2-minute processing on a 2 million line mesh file (in our testing, your milage may vary.)
- **3D Visualisation**: Interactive web-based material distribution viewer
- **Mesh Transformation**: Built-in rotation and translation for CT alignment
- **Advanced Integration**: Multiple integration schemes with adaptive accuracy
- **Wide Format Support**: Abaqus (.inp), ANSYS (.cdb), VTK, and DICOM formats
- **Parallel Processing**: Multi-threaded material assignment with progress reporting

Installation

For more information on installation, please see the Installation Guide.

Easy install

The compiled installers are available in the release

Manual Build Commands

```
cargo build --release
```

Quick Start

See the commands with LUMA's help flag:

```
luma -h
```

Basic Material Assignment

```
luma --params example_parameters.toml --ct example_ct_data.vtk --mesh
example_mesh.inp
```

3D Visualisation

```
luma --visualise align --ct scan.vtk --mesh bone.inp
```

Material Assignment Visualisation

```
luma --visualise processed --params parameters.toml --ct scan.vtk --mesh bone.inp
```

With Mesh Transformation

```
luma --params parameters.toml --ct scan.vtk --mesh bone.cdb --rot 180.0 0.0 0.0 --
trans 0.0 0.0 10.0
```

Features

Material Assignment

The core functionality assigns material properties to finite element meshes based on CT scan data through a multi-step pipeline:

1. **CT Data Processing**: Load CT data from VTK or DICOM files
2. **Mesh Loading**: Parse Abaqus (.inp), ANSYS (.cdb) or VTK (.vtk/.vtu) mesh files
3. **Spatial Integration**: Sample CT values within each finite element
4. **Material Property Calculation**: Convert Hounsfield units to material modulus
5. **Material Grouping**: Group similar elements into material sets
6. **Output Generation**: Write material assignments to output files

Development

Project Structure

```
frontend/
├── components/             # Javascript components for UI functionality
├── styling/                # CSS files for UI styling
├── favicon.ico             # Icon
└── index.html              # Main html script for creating UI
src/
├── lib.rs                  # Main library interface
├── main.rs                 # Command line application
├── export/                 # Output file generation
│   ├── abaqus_inp.rs       # Abaqus .inp output
│   ├── ansys_cdb.rs        # ANSYS .cdb output
│   └── model_vtk.rs        # VTK model file output
├── mesh/                   # Mesh file parsing
│   ├── abaqus_inp.rs       # Abaqus .inp parser
│   ├── ansys_cdb.rs        # ANSYS .cdb parser
│   └── model_vtk.rs        # VTK model file parser
├── integrate/              # Integration methods
├── params/                 # Parameter file parsing
├── visualise/              # 3D visualisation
│   ├── mesh_visualiser.rs  # Creates visualisation
│   └── web_viewer.rs       # Spawns UI for visualisation
├── volume/                 # CT data handling
│   ├── image_dicom.rs      # DICOM image parsing
│   └── image_vtk.rs        # VTK image parsing
│   └── image_nifti.rs      # NIfTI image parsing
```

Disclaimer

This software has been designed for research purposes only and has not been reviewed or approved by medical device regulation bodies.

This software is not to be used alone or in combination, for human beings for one or more of the following specific medical purposes:

- diagnosis, prevention, monitoring, prediction, prognosis, treatment or alleviation of disease.
- diagnosis, monitoring, treatment, alleviation of, or compensation for, an injury or disability.
- investigation, replacement or modification of the anatomy or of a physiological or pathological process or state.
- providing information by means of *in vitro* examination of specimens derived from the human body, including organ, blood and tissue donations.

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Credit

This project was inspired by py_bonemat_abaqus by Elise Pegg currently of Newcastle University and Richie Gill currently of University of Bath. And the work of Istituto Ortopedico Rizzoli in Bologna, Italy, particularly the work of the Bonemat and ALBA teams.

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